ANKARA UNIVERSITY

FACULTY OF ENGINEERING

Departments of Computer Engineering, Biomedical Engineering, Electrical and Electronics Engineering, and Software Engineering

MULTIDISCIPLINARY PROJECT STUDY

SPRING SEMESTER 2025–2026

Project Title: Design and Testing of an Audiometer System

Project Aim

An audiometer is an electronic medical device used to measure an individual's hearing sensitivity. The device delivers pure tones to the patient through headphones at different frequencies and intensity levels, typically ranging from 250 Hz to 8000 Hz, and determines the hearing thresholds of each ear separately. The patient indicates when they hear a sound by pressing a button, and the obtained data are displayed on an audiogram graph in decibels (dB). This method allows clinicians to determine the presence, degree, and frequency range of hearing loss.

The aim of this project is to develop a hearing test device (audiometer) that complies with medical standards. Four teams, consisting of students from Computer Engineering, Biomedical Engineering, Electrical and Electronics Engineering, and Software Engineering, will collaboratively design, implement, and test this system through an interdisciplinary approach.

Project Groups

Each group will be formed to include one team from each of the four departments listed below. For each group, one team leader will be assigned from each department, and one overall group leader will be selected from among these team leaders.

Department	**Course Code**	**Course Name**
Computer Engineering	COM 2044	Object Oriented Programming
Biomedical Engineering	BME 324 BME 4400	Biomedical Instrumentation Internship
Electrical and Electronics Engineering	EEE3002	Electronics II
Software Engineering	YMH 334	Fonksiyonel Programlama

Contact information will be entered via the link below. (Groups will be formed randomly.)

https://docs.google.com/spreadsheets/d/1Nlk44StMFXDcDbcTX2ZQfFT_2kUi5d3TSpmOniYa_Dw/edit?usp=sharing

System Architecture

Virtual Hardware (Proteus):

- It consists of an Arduino UNO, COMPIM module, MCP4921 DAC, and an LM358 Op-Amp buffer circuit. It generates medical-grade pure sine waves. The patient's button response is simulated using a virtual button in the Proteus interface; when clicked, a "RESPONSE" message is sent to the Java application.

Clinical Software (Java):

- This is the application where the audiologist manages the test, the Hughson–Westlake algorithm is executed, and the audiogram is drawn in real time.

Physical Button [Optional]

Although the entire system operates in a virtual environment, two alternative hardware platforms will be provided upon request for groups who wish to gain hands-on experience with a physical implementation. These platforms are used solely to physically represent the patient response button; sound generation and circuit design remain in the virtual environment.

The ESP32-S3 is a microcontroller (MCU)-based platform that integrates the processor, memory, and peripherals on a single chip and is programmed using C/C++. The Tang Nano 9K, on the other hand, is an FPGA (Field-Programmable Gate Array) board; instead of software, hardware logic is programmed, operations are executed in parallel, and VHDL or Verilog is used.

Since both platforms directly support serial communication over USB, pressing the button sends a "RESPONSE" message to the Java application. Therefore, no changes are required in the software architecture.

	Proteus (Default)	ESP32-S3 (Option A)	Tang Nano 9K — FPGA (Option B)
Button	Virtual (on screen)	Physical	Physical
Programming	—	C/C++	VHDL / Verilog
Java	—	-	-
Voice generation	Virtual	Virtual	Virtual

Task Distribution Among Teams

Electrical and Electronics Engineering Team

- Set up the Arduino UNO, COMPIM, MCP4921 DAC, and LM358 buffer circuit in Proteus
- Generate pure sine waves via the DAC based on commands received from Java (C/C++)
- Configure the virtual button; if an optional physical button is selected, ensure serial port integration

Computer Engineering Team

- Design a GUI using Java (Swing / JavaFX) for test management and frequency/intensity selection
- Send commands to the system and read "RESPONSE" messages via jSerialComm
- Plot threshold values on a standard audiogram (right ear: red O, left ear: blue X)

Biomedical Engineering Team

- Document the Hughson–Westlake procedure ("down 10 dB, up 5 dB") as a rule set and deliver it to other teams; define the test frequency range (250 Hz – 8000 Hz)
- Verify that the signal is a clean "pure tone" using THD analysis with the Proteus oscilloscope and spectrum analyzer
- Explain the IEC 60645-1 standard in the report
- Provide information on the working principles of the audiometer and test procedures

Software Engineering Team

- Implement all medical computations as pure functions
- Manage test data using immutable data structures
- Process RESPONSE messages using map / filter / reduce chains
- Handle error cases without side effects using the Optional / Maybe pattern
- Verify compliance with IEC 60645-1 algorithm rules defined by the biomedical team by writing automated unit tests and property-based tests

Project Report Requirements

- The report must not exceed 15 pages, excluding code and appendices
- All references and standards used must be clearly stated in the report
- Direct quotations from the assignment text are not allowed; all figures and diagrams must be original
- Each group must hold at least three in-person meetings during the project; signed meeting minutes must be included in the appendix
- The report must be completed and submitted via the e-campus/Google Classroom system in the specified format before the final exam

TABLE OF CONTENTS

The pdf formatted reports should be uploaded to course ekampus page.

One Report from team leaders should be uploaded to ekampus / Google Classroom page.

Ankara University, Faculty of Engineering, Department of Biomedical, Computer, Sand Electrical and Electronics Engineering	MULTIDISCIPLINARY PROJECT WORK EVALUATION REPORT	Semester: 2025 – 2026 Spring Term

TEAM

Number		**Name**	

STUDENTS

Number	**Name and Surname**	**Role in Group (Not contributed, Member, Team leader, Group leader)**	**Course Code**

Evaluation		Weights	Grade
1	Literature Review	10	
2	Project Management	10	
3	System Design and Implementation (Hardware OR Simulation)	10	
4	Signal Generation	10	
5	Software Design	10	
6	Implementation of Hughson-Westlake Algorithm	10	
7	System Integration and End-to-End Testing	10	
8	Signal Analysis and Validation	10	
9	Standards and Engineering Analysis	10	
10	Report Quality and Multidisciplinary Collaboration	10	
	Total	100	

Evaluator(s):